

Differential Calculus — Logarithmic Differentiation

Worked Problem 03

Chapter 2: Rational-Inverse Trigonometric-Exponential Function | Page 34

PROBLEM STATEMENT

Logarithmic Differentiation of Complex Quotient

Find $\frac{dy}{dx}$ from the given multi-composite function:

$$y = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$$

STEP 1 প্রদত্ত জটিল ভগ্নাংশীয় ফাংশনটি উল্লেখ করি (Given Function)

$$y = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$$

STEP 2 উভয়পক্ষে স্বাভাবিক লগারিদম (log) নিয়ে গুণ ও ভাগকে যোগ-বিয়োগে রূপান্তরিত করি (Applying Logarithmic Laws for Products & Quotients)

$$\log y = \log(e^{x^2}) + \log(\tan^{-1} x) - \log(\sqrt{1+x^2}) = x^2 + \log(\tan^{-1} x) - \frac{1}{2} \log(1+x^2)$$

STEP 3 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Differentiating Both Sides w.r.t. x)

$$\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx}(x^2) + \frac{d}{dx} [\log(\tan^{-1} x)] - \frac{1}{2} \frac{d}{dx} [\log(1+x^2)]$$

STEP 4 বিপরীত ত্রিকোণমিতিক ও সূচকীয় সূত্রের চেইন রুল প্রয়োগ করি (Applying Chain Rule to Each Component)

$$\frac{1}{y} \frac{dy}{dx} = 2x + \frac{1}{\tan^{-1} x} \cdot \frac{1}{1+x^2} - \frac{1}{2(1+x^2)} \cdot (2x) = 2x + \frac{1}{(1+x^2) \tan^{-1} x} - \frac{x}{1+x^2}$$

STEP 5 উভয়পক্ষকে y -এর মান দ্বারা গুণ করে চূড়ান্ত অন্তরজ পাই (Multiplying by y for Final Derivative)

$$\frac{dy}{dx} = y \left[2x + \frac{1}{(1+x^2) \tan^{-1} x} - \frac{x}{1+x^2} \right] = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}} \left(2x + \frac{1}{(1+x^2) \tan^{-1} x} - \frac{x}{1+x^2} \right)$$

FINAL ANSWER
$$\frac{dy}{dx} = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}} \left(2x + \frac{1}{(1+x^2) \tan^{-1} x} - \frac{x}{1+x^2} \right)$$

■ BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- **Logarithmic Transformation:** $\log\left(\frac{u \cdot v}{w}\right) = \log u + \log v - \log w$.
- **Standard Inverse Tangent Derivative:** $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$.
- **Log of Inverse Trig Composite:** $\frac{d}{dx}[\log(\tan^{-1} x)] = \frac{1}{\tan^{-1} x} \cdot \frac{1}{1+x^2} = \frac{1}{(1+x^2)\tan^{-1} x}$.
- **Log of Quadratic Radical:** $\frac{d}{dx}\left[\frac{1}{2}\log(1+x^2)\right] = \frac{1}{2} \frac{2x}{1+x^2} = \frac{x}{1+x^2}$.

⚠ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Forgetting Coefficient $\frac{1}{2}$ on Radical Denominator:** Differentiating $\log \sqrt{1+x^2}$ without pulling the $\frac{1}{2}$ exponent forward leads to an incorrect $2x/(1+x^2)$.
- **Denominator Minus Sign Error:** Because the term $\sqrt{1+x^2}$ is in the denominator, its logarithmic sign is $-\frac{1}{2}\log(1+x^2)$.
- **Unsimplified Direct Quotient Rule Trap:** Directly applying $\frac{vu'-uv'}{v^2}$ results in extremely unwieldy algebra prone to multiple arithmetic mistakes.

🌐 BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Inverse Tangent Positivity:** For $\log(\tan^{-1} x)$ to be real-valued, $\tan^{-1} x > 0 \implies x > 0$.
- **Denominator Radical:** $\sqrt{1+x^2} > 0$ for all $x \in \mathbb{R}$.
- **Strict Real Domain:** Domain = $(0, \infty)$.
- **Smoothness:** Infinitely differentiable and analytic on $(0, \infty)$.

💡 BOX 4: DIRECT TRIPLE PRODUCT FORMULATION VERIFICATION

- Rewriting as product: $y = e^{x^2} \cdot \tan^{-1} x \cdot (1+x^2)^{-1/2}$
- Using general product rule $(uvw)' = u'vw + uv'w + uvw'$:

$$\frac{dy}{dx} = (2xe^{x^2})\tan^{-1} x(1+x^2)^{-1/2} + e^{x^2}\left(\frac{1}{1+x^2}\right)(1+x^2)^{-1/2} + e^{x^2}\tan^{-1} x\left(-\frac{1}{2}(1+x^2)^{-3/2} \cdot 2x\right)$$

- Factoring out $y = \frac{e^{x^2}\tan^{-1} x}{\sqrt{1+x^2}}$ instantly reproduces the exact same expression in brackets!

🎯 BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- **Step 1–2 (Logarithmic Transformation of Quotient):** 1.0 Mark
- **Step 3 (Implicit Logarithmic Differentiation Operator):** 0.5 Mark
- **Step 4 (Inverse Trig & Quadratic Chain Derivatives):** 1.5 Marks
- **Step 5 & Final Expressive Form:** 1.0 Mark
- **Total Allocation:** 4.0 Marks